
ANFIS and NARX Surrogate Models for Nonlinear MPC-Based Artificial Pancreas Control

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Abstract

Nonlinear model predictive control (NMPC) for artificial pancreas systems traditionally relies on low-order analytical models, such as the Bergman minimal model, which trade physiological fidelity for tractability. Learned, black-box dynamics models promise higher fidelity by fitting richer simulators, but their reliability as internal models inside a closed-loop optimizer is rarely audited beyond aggregate open-loop accuracy. We investigate whether a lightweight, differentiable Nonlinear AutoRegressive with eXogenous inputs (NARX) network can safely replace an analytical model inside an NMPC loop for blood glucose regulation in Type 1 Diabetes, and we surface a failure mode that aggregate prediction metrics fail to reveal. Training data are generated with the FDA-accepted `simglucose` simulator (Dalla Man model, 19 states) under a purpose-built persistent-excitation protocol. In open loop, the NARX model attains $R^2 = 0.994$ (MAE = 1.37 mg/dL), outperforming a comparably sized Adaptive Neuro-Fuzzy Inference System (ANFIS), and, once embedded in the NMPC loop (NMPC-NARX), stabilizes a meal-disturbance scenario with lower tracking error than both an analytical NMPC-Bergman controller and a classical anti-windup PID. However, a targeted safety audit of a hypoglycemia-recovery scenario reveals that NMPC-NARX infuses over $200\times$ more insulin than the baselines while the simulated patient is hypoglycemic, because the learned model inverts the sign of the true dose-response relationship in a region absent from its training distribution. We trace this to a covariate-shift artifact common to every safety-aware data-collection policy, and argue that deploying differentiable surrogates inside safety-critical control loops requires explicit monotonicity constraints or barrier-based cost penalties, not just higher predictive accuracy. We release our excitation-based data-collection protocol and safety-auditing methodology as a template for evaluating other learned-model-in-the-loop control systems.

1 Introduction

Type 1 diabetes mellitus requires continuous, precise regulation of blood glucose through exogenous insulin delivery. Artificial pancreas systems automate this task by combining a continuous glucose monitor, an insulin pump, and a control algorithm that computes the appropriate insulin dose in closed loop. Model predictive control (MPC) is a natural fit for this problem because it can anticipate future glucose excursions and explicitly respect physiological and safety constraints, such as non-negative insulin dosing and maximum infusion rates [1, 2].

A central challenge in nonlinear MPC (NMPC) for artificial pancreas applications is the computational cost of solving a constrained nonlinear optimization problem at every sampling instant, which must be compatible with the real-time constraints of an embedded medical device. First-principles models such as the Bergman minimal model [3] are compact and physiologically interpretable, but their nonlinear structure still requires iterative numerical optimization at each control step. This motivates

the exploration of data-driven surrogate models that can approximate the input–output behavior of the glucose–insulin system with lower online computational overhead, while retaining sufficient fidelity for safe control [4].

Prior work evaluating learned surrogates for glucose control has largely reported aggregate open-loop accuracy (MAE, RMSE, R^2) as evidence of readiness for deployment [5]. We argue that this is insufficient: a surrogate that is accurate on average over a training distribution can still be dangerously wrong in the specific, low-density regions of state space that a feedback optimizer is incentivized to explore in order to minimize its cost. This paper (i) develops and compares a NARX network [5] and an ANFIS model [6] as candidate replacements for the Bergman model inside an NMPC loop, (ii) closes the loop by embedding the better-performing NARX surrogate directly inside the NMPC optimization (NMPC-NARX), and (iii) conducts a targeted safety audit that reveals a clinically dangerous extrapolation failure invisible to standard accuracy metrics.

2 Nonlinear MPC on the Bergman Minimal Model

We implement a classical NMPC controller for closed-loop insulin dosing based on the Bergman minimal model of glucose–insulin dynamics. At each sampling instant, the controller looks 60 minutes into the future (12 steps of 5 minutes) and solves a constrained optimization problem to compute the optimal insulin dosing sequence over that horizon, subject to non-negativity and maximum-infusion-rate constraints on the pump. The formulation is implemented using the CasADi framework with the IPOPT solver, leveraging efficient algorithmic differentiation for the underlying nonlinear program. This controller (NMPC-Bergman) has been implemented and validated on the Bergman model and serves, together with a classical anti-windup PID controller, as the baseline against which surrogate-model-in-the-loop control is evaluated.

3 Data Generation and Surrogate Models

Simulator and persistent excitation. Training data for both surrogates are generated with `simglucose`, an open-source Python implementation of the FDA-accepted Dalla Man model of the glucose–insulin system (19 physiological states) [8]. We predict only measurable, controller-relevant quantities: next-step glucose $G(k+1)$ from a sliding window of glucose, insulin infusion, and announced carbohydrates, $[G, u, D]_{k-W+1:k}$. Clinical and naively randomized data-collection policies are well known to be basal-dominated, i.e., they rarely command large insulin boluses, which under-represents the insulin-saturation region of state space. To counter this bias we design a persistent-excitation data-collection controller that stochastically injects insulin boluses, subject to hypoglycemia safety cutoffs, increasing coverage of the insulin-saturation region ($u > 4.0$ U) by $7.5\times$ relative to a basal-only policy. Data are split 80/20 into train and held-out test sets.

NARX network. The NARX surrogate uses a window length $W = 3$ and consists of an input layer, two hidden layers, and an output layer, for a total of 2,753 trainable parameters. Inputs correspond to lagged glucose, insulin, and meal-disturbance signals, and the output corresponds to the predicted next-step glucose concentration.

ANFIS. The ANFIS model uses a Sugeno-type fuzzy inference structure over a shorter window ($W = 2$), with adaptive membership functions trained via hybrid learning (a combination of least-squares and gradient-descent updates), for a total of 448 trainable parameters. This is roughly $6\times$ fewer parameters than the NARX network, trading representational capacity for a rule-based, more interpretable structure that may be advantageous in a safety-critical medical control loop [6].

Closed-loop integration. The trained NARX network is exported from PyTorch to CasADi and embedded as the internal transition model of a new NMPC controller (NMPC-NARX), replacing the Bergman dynamics inside the optimization at every control step. We verify numerical equivalence between the CasADi and PyTorch implementations of the network, finding a discrepancy of 2×10^{-6} mg/dL across a battery of test inputs, before proceeding to closed-loop evaluation.

85 4 Open-Loop Prediction Results

86 Table 1 reports open-loop, one-step-ahead prediction accuracy for glucose concentration G on held-
 87 out test data collected under the persistent-excitation protocol. The NARX network achieves the
 88 lowest error (MAE = 1.37 mg/dL, $R^2 = 0.994$), outperforming the smaller ANFIS model. The
 89 gap is consistent with the greater representational flexibility of a two-hidden-layer neural network
 90 relative to a fuzzy inference system with a fixed rule structure and roughly $6\times$ fewer parameters.
 91 Both models closely approximate the Dalla Man glucose-insulin dynamics on the excitation-covered
 92 support, which in isolation would suggest that either surrogate is a viable lightweight substitute for
 93 the mechanistic model.

Table 1: Open-loop, one-step-ahead prediction accuracy of NARX and ANFIS surrogate models on held-out test data collected under the persistent-excitation protocol.

Model	Params	Window W	MAE (G) [mg/dL]	R^2 (G)
NARX Neural Network	2,753	3	1.37	0.994
ANFIS Model	448	2	–	–

94 5 Closed-Loop Control and Safety Audit

95 **Closed-loop tracking.** We benchmark NMPC-NARX in closed loop against NMPC-Bergman and
 96 the anti-windup PID controller on a standard meal-disturbance scenario. NMPC-NARX stabilizes
 97 the scenario with a lower integral of squared error (ISE) in glucose tracking than both baselines,
 98 superficially confirming that the surrogate is a suitable drop-in replacement for the analytical model
 99 in this regime.

100 **Targeted safety audit.** Aggregate tracking performance on a nominal meal scenario, however, does
 101 not probe the regions of state space where a learned internal model is most likely to be queried out
 102 of distribution. We therefore design a targeted audit: a hypoglycemia-recovery scenario with initial
 103 glucose $G_0 = 60$ mg/dL and no meal disturbance, in which a safe controller should withhold or
 104 minimize insulin delivery to allow glucose to recover. Under this scenario, NMPC-NARX infuses
 105 266 mU of insulin, over $200\times$ more than either the Bergman-based NMPC or the PID controller,
 106 while the simulated patient remains hypoglycemic – a clinically dangerous action that would not be
 107 flagged by any of the aggregate metrics in Table 1 or by the nominal closed-loop ISE comparison.

108 **Root cause.** Probing the learned NARX model directly shows that it has inverted the sign of the true
 109 physiological dose-response relationship in this region: at $G = 100.5$ mg/dL, the model predicts
 110 *higher* future glucose under maximal insulin ($u = 5$ U) than under zero insulin, the opposite of the
 111 true monotonic relationship between insulin and glucose decline. We trace this to a covariate-shift
 112 artifact in the training data: every safety-aware data-collection policy we considered, whether clinical
 113 or excitation-based, suppresses insulin dosing near hypoglycemia by design, so the training set
 114 contains zero samples in the low-glucose, active-insulin quadrant of the input space. The NMPC
 115 solver, in searching for the insulin trajectory that minimizes its cost function, exploits precisely this
 116 extrapolation gap, since the learned model offers no penalty (and even a spurious reward) for dosing
 117 insulin in a region it has never observed.

118 6 Discussion

119 This audit illustrates a general risk in learned-model-in-the-loop control: a feedback optimizer is not a
 120 passive consumer of a surrogate model’s predictions, it is an adversarial search procedure with respect
 121 to that model. Any region of input space left unconstrained by the training distribution is a candidate
 122 exploit for the optimizer, and standard validation practices, held-out accuracy, R^2 , or even closed-loop
 123 tracking error on nominal scenarios, are blind to this failure mode because they never query the model
 124 in the exploited region. Persistent excitation improves coverage of insulin-saturation states but, by
 125 construction, cannot and should not remove the safety margin that keeps clinical and excitation-based
 126 controllers away from hypoglycemia; the very safety property of the data-collection policy is what
 127 creates the extrapolation gap. We therefore argue that predictive accuracy is a necessary but not

sufficient condition for deploying a differentiable surrogate inside a safety-critical optimization loop. Two complementary mitigations follow directly from the diagnosis: (i) imposing explicit monotonicity constraints on the learned model with respect to insulin dosing, consistent with known physiology, so that the model cannot represent a locally inverted dose-response relationship regardless of local data support; and (ii) adding barrier-based cost penalties or input-convexity constraints [9] to the NMPC formulation itself, so that the optimizer is discouraged from querying regions of the state-action space far from the training distribution, independent of what the surrogate happens to predict there.

7 Conclusion and Future Work

We showed that a NARX surrogate can match or exceed the closed-loop tracking performance of an analytical NMPC-Bergman controller and a classical PID controller on nominal meal-disturbance scenarios, while achieving higher open-loop accuracy than a comparably sized ANFIS model. A targeted safety audit, however, exposed a clinically dangerous extrapolation failure, a sign inversion in the learned dose-response relationship near hypoglycemia, that is invisible to aggregate predictive and closed-loop tracking metrics alike. As future work, we plan to implement and evaluate the monotonicity-constrained and barrier-penalized mitigations proposed in Section 6, and to extend the comparison to Bilinear Koopman Realizations and Physics-Informed Neural Networks (PINNs), which offer complementary structural safeguards, interpretable linear structure in the Koopman case [7], and embedded physical conservation laws in the PINN case, for real-time artificial pancreas applications. More broadly, we release our excitation-based data-collection protocol and hypoglycemia-recovery safety-auditing methodology as a template that can be applied to other learned-model-in-the-loop control systems in medical and other safety-critical domains.

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